

FINAL TECHNICAL REPORT

Exactness at Scale: From CC(8) to 451 Million Determinants in Rust

Quantum Harness Challenge #129 / Team Ranger
Chenxi Wan, Yedi Shen, Junkai Wang
Innovation-led public delivery / 2026-07-30

PUBLISHED ACCURACY	LARGEST EXACT SECTOR	WALL TIME	HPC CAMPAIGN
36/36	451,681,246	3:55:43	560 CPUs

Breakthrough

Ranger delivers one Rust determinant engine for FCI, CC, CI, MBPT, UCC, direct integrals, symmetry resolution, deterministic parallel execution, restartable Davidson, and verified HPC evidence. Exact ranked subset convolution makes CC(1)-CC(8) practical; symmetry-compact matrix-free FCI carries the same engine to an exact 451,681,246-determinant sector.

The result is a complete path from equations to algorithms, from algorithms to hundreds of millions of determinants, and from one benchmark to a reusable platform for selected electronic-structure methods.

1. Three barriers, three algorithms

1 / Wave-function construction

Exact ranked subset convolution builds $\exp(T)|HF\rangle$ by excitation rank. Each target coefficient combines compatible amplitude/source partitions with exact fermionic phases and reuses completed lower-rank coefficients. A Taylor implementation supplies an independent coefficient oracle.

2 / Hamiltonian scale

Matrix-free spin-free sigma applies the Hamiltonian through signed string links. Compact ORBSYM/ISYM addresses propagate across FCI, CI, MBPT, CC, and UCC, transforming 1,806,590,016 determinants into 451,681,246 exact C2v/A1 determinants for the water ground state.

3 / Reproducible production

Fixed source blocks, thread-local partial vectors, and ordered reduction make a fixed parallel policy bitwise repeatable. Versioned vector stores, atomic checkpoint generations, input hashes, memory preflight, and resume turn long Davidson runs into durable scientific computations.

Why the advances compound

Advance	Scaling effect	Measured proof
Ranked subset convolution	Direct finite-rank CC wave function	CC(1)-CC(8), 8/8
Matrix-free sigma	Operator storage replaces matrix storage	28M and 451M sectors
Compact symmetry addresses	Fourfold exact representation gain	1.806B -> 451M
Deterministic Davidson	Parallel, repeatable, restartable	3.236817x + HPC

2. Primary acceptance: 36/36

The primary H2O/6-31G Hamiltonian freezes the oxygen 1s orbital and contains 12 active spatial orbitals, eight active electrons, and 245,025 determinants. Every published equilibrium entry matches at the six decimal places printed by Hirata and Bartlett.

Method family	Range	Matches	Representative value
Coupled cluster	CC(1)-CC(8)	8/8	CC(8) -76.121174196144139 Eh
Configuration interaction	CI(1)-CI(8)	8/8	CI(8) reaches FCI
Perturbation theory	MBPT(1)-MBPT(20)	20/20	All printed orders
Matrix-free FCI	Ground state	Accepted	-76.121174204141965 Eh

Independent numerical ladder

Dense Rust FCI agrees with PySCF for H2, linear H4, and H2O/STO-3G. The direct, serial, parallel, symmetry, memory, disk, checkpoint, resume, multi-root, and full-rank UCC paths cross-check shared algebra through distinct executions. Stretched-water fixtures at 1.5 and 2.0 times the equilibrium bond length extend acceptance across changing correlation regimes.

High-rank coupled cluster

CC(2) agrees with the independent PySCF CCSD oracle within 3.025e-10 Eh. CC(8) reaches 7.998e-9 Eh from FCI, while CI(8) reaches 2.004e-12 Eh. The complete CC series finishes in 186.94 seconds on the recorded Apple M4 environment, turning general-order theory into a practical test oracle.

3. Exact scaling ladder

Hamiltonian and exact sector	Determinants	Rust energy
H2O/6-31G, O 1s frozen	245,025	-76.121174204142 Eh
H2O/DZ, all electron	1,002,708	-76.156699030930056 Eh
H2O/DZP, O 1s frozen	28,233,466	-76.256624441300147 Eh
H2O/cc-pVDZ, all electron, C2v/A1	451,681,246	-76.24321859 Eh

451-million-determinant exact sector

The largest run correlates all ten electrons in 24 spherical spatial orbitals. Davidson reaches residual 6.602e-08 in 21 iterations and 03:55:43 wall time. Same-input PySCF RHF, MP2, CISD, CCSD, and CCSD(T) supply a method hierarchy; CCSD(T) lies 0.647144 mEh above the Rust FCI result.

Symmetry-free resource characterization

The companion 1,806,590,016-determinant study measures a 13.460145 GiB CI vector, Rust integral generation, RHF, AO-to-MO transformation, string links, distributed source samples, and sparse Hamiltonian columns. The fourfold compact A1 representation connects this resource model to the completed production solve.

Scale multiplier

The largest completed exact sector contains more than 1,800 times the determinants of the primary challenge. Rank recursion, matrix-free action, symmetry addressing, parallel reduction, and restartable storage create this multiplier together.

4. Deterministic HPC

The SCNet workflow rebuilds the pinned Rust source in an offline toolchain, runs the test suite and numerical smoke checks, sweeps Davidson parameters, and repeats solves across AMD EPYC 7742 nodes.

Evidence	Result
Robustness matrix	18/18 converged
Repeated solves	216/216 converged
Energy range	8.10e-13 Eh
Verified manifests / evidence files	37 / 960
Allocation peak	560 CPUs across 10 tasks

Local kernel acceleration

Median sigma time moves from 14.181091542 seconds serial to 4.381184834 seconds with four fixed blocks, a 3.236817x ratio of medians. Ordered block reduction yields bitwise repeatability for a fixed policy and links the local kernel test directly to production execution.

Throughput architecture

Per-solve utilization measurements motivate four 14-thread solver processes per 56-core node. The 1,008-CPU campaign design schedules 72 independent processes across 18 nodes, providing a concrete architecture for high-volume determinant method studies. The verified 560-CPU campaign establishes the repeatability and portability foundation for this design.

5. One engine, many methods

The workbench is organized around reusable determinant algebra. Direct libcint AO integrals flow through Rust RHF/DIIS and staged AO-to-MO transformation into the same determinant basis, symmetry addresses, and Hamiltonian action used by every wave-function method.

Shared primitive	Method families that reuse it
Determinant addresses + fermionic phases	FCI, CI, CC, MBPT, UCC
Matrix-free Hamiltonian action	FCI, projected CI, CC residuals
Compact symmetry sectors	FCI, CI, MBPT, CC, UCC
Davidson vector stores	Ground states, several roots, selected spaces
Machine-readable fixtures	Dense, direct, parallel, HPC acceptance

Selected-determinant frontier

The exact engine prepares a common interface for deterministic HCI/iCI-style selection, variational selected-space Davidson, Epstein-Nesbet PT2, threshold extrapolation, natural-orbital iterations, orbital optimization, and quantum-sampled determinant lists. The completed exact result ladder supplies calibration targets and equal-size comparisons for every selection strategy.

Promising research direction

Ranger combines the accuracy of an exact finite-sector oracle with an architecture built for method invention. New determinant generators can enter through one address layer and immediately inherit symmetry, matrix elements, eigensolvers, fixtures, residual checks, and reproducible evidence.

6. Reproduction and public package

Complete local acceptance

```
uv sync --locked
scripts/verify-submission.sh
```

Focused evidence audit

```
python3 scripts/hpc/verify_final_evidence.py
```

Artifact	Repository path
Technical article	reports/final-competition-summary.md
Technical PDF	output/pdf/quantum-harness-129-final-technical-report.pdf
Result card	output/data/quantum-harness-129-final-results.txt
Checksum manifest	output/quantum-harness-129-submission-manifest.txt
451M machine record	fixtures/h2o-ccpvdz-ae/fci-c2v-xh5-result.json
SCNet machine record	fixtures/hpc/scnet-2026-07-30.json
Reproduction prompt	docs/reproduction-prompt.md

Public endpoints

Repository: <https://github.com/JunkaiWang-TheoPhy/quantum-harness-129-workbench-rust>
Publication branch: `codex/final-competition-submission`
PR: <https://github.com/QuantumBFS/quantum.harness/pull/217>
Release: <https://github.com/JunkaiWang-TheoPhy/quantum-harness-129-workbench-rust/releases/tag/v0.5.0>

Scientific foundations

Hirata-Bartlett general-order CC: DOI 10.1016/S0009-2614(00)00387-0
Knowles-Handy determinant FCI: DOI 10.1016/0009-2614(84)85513-X
Holmes-Tubman-Umrigar HCI: DOI 10.1021/acs.jctc.6b00407
Zhang-Liu-Hoffmann iCI: DOI 10.1021/acs.jctc.9b01200

Final perspective

Ranger demonstrates a powerful path for electronic-structure research: build one trusted determinant algebra, introduce algorithms whose gains compound, and carry every advance from equations to source, data, HPC evidence, and public reproduction.